Andrés Gallardo Agüero

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Profile

Economics master's student at the University of St.Gallen working at the intersection of causal machine learning and clinical data. Developing a master's thesis on variable importance for treatment effect heterogeneity, and co-authoring a paper applying causal forests to ICU data accepted for presentation at two international conferences. Strong background in econometrics, causal inference, and statistical learning, with applied research experience using large-scale real-world datasets in clinical, macroeconomic, and financial settings.

Research Interests

Causal machine learning and heterogeneous treatment effects; methods for individualized clinical decision-making; applied econometrics and causal inference.

Education

University of St.Gallen **Sep 2024 to Feb 2027**

Master of Arts in Economics, School of Economics and Political Science *St. Gallen, Switzerland*

- Specialization in Data Science and AI in Economics.
- Vice President of the Ad-Hoc Economics Club. Member of ACM HSG and the Healthcare Club.

Universidad Adolfo Ibáñez **Mar 2020 to Dec 2023**

Bachelor of Science in Economics, School of Business *Santiago, Chile*

- Final GPA 6.2 out of 7.0. Ranked 9 out of 108 graduating students.
- Pedro Luis González Academic Excellence Scholarship (2020 to 2023).
- Academic Honor Scholarship (2021 to 2023).

Research

Master's Thesis (in progress)

Do Causal Forest Methods Identify the Right Sources of Treatment Effect Heterogeneity? Evidence from Clinical Data with Known Ground Truth

- Supervised by Dr. Justus Vogel; co-supervised by Prof. Dr. Jana Mareckova.
- Simulation study with known ground truth comparing whether variable importance measures correctly identify the variables that drive heterogeneous treatment effects, and how reliably causal forest estimators (GRF, MCF) recover those effects, using clinical data generating processes.

Working Paper (in progress)

Whom to prioritize? Causal effects of treatment target non-adherence on mortality in intensive care medicine

- Authors: Justus Vogel, Julian Klug, Andrés Gallardo, Miodrag Filipovic, Urs Pietsch.
- Estimates causal effects of mean arterial pressure target non-adherence on 28-day ICU mortality using generalized random forests, with individualized effects to support patient prioritization.
- Accepted for presentation at the EuHEA Conference 2026 and the ASSA 2027 Annual Meeting (presented by co-author Justus Vogel).

Research Experience

University of St.Gallen **Jan 2025 to Present**
Research Assistant, School of Medicine *St. Gallen, Switzerland*

- Lead the empirical implementation of a causal inference project on a large-scale ICU dataset.
- Estimate heterogeneous treatment effects using causal forests in R to support individualized clinical decision-making in critical care settings.

University of St.Gallen **Mar 2023 to Aug 2024**
Research Assistant, Institute of Economic Research and Policy *Remote*

- Conducted empirical macroeconomic analysis using economic and financial time series data.
- Applied the synthetic control method to evaluate intervention impacts and inform policy decisions.

Universidad del Desarrollo **May 2024 to Aug 2024**
Research Assistant *Remote*

- Conducted empirical microeconomic analysis using nationally representative survey data.
- Applied difference-in-differences methods to estimate treatment effects across cohorts.

Central Bank of Chile **Jan 2024 to Feb 2024**
Research Assistant Intern, Monetary Policy Division *Santiago, Chile*

- Supported a research project on Growth-at-Risk methodology by constructing a financial conditions index for the Chilean economy.
- Prepared datasets and analytical outputs to support internal research and monetary policy discussions.

Universidad Adolfo Ibáñez **Mar 2021 to Aug 2024**
Research and Teaching Assistant *Hybrid*

- Research assistant across multiple empirical projects in macroeconomics, microeconomics, and finance.
- Teaching assistant for courses in economics, statistics, and econometrics.

Other Experience

Pacifico Research **Jul 2025 to Sep 2025**
Quantitative Research Intern *Remote*

- Modeled the Chilean sovereign yield curve using bond-level time series data in Python.
- Identified relative value opportunities using machine learning to inform fixed income allocation strategies.
- Performed data cleaning, feature construction, and validation on fixed income time series data.

Additional Information

Programming and Tools: R, Python, LaTeX, Git, STATA, Julia, SQL.
Languages: Spanish (native), English (C2, TOEFL iBT 114/120, 2023), German (A1).
Citizenship: Chilean.